

# FYP 2

## Chapter 4

### Design and Implementation

#### Chapter 4 : Design

- In requirement phase, you have developed flowchart, use case model etc
- In this chapter you will design the system/tools/application –

Refer the flowchart and use case model from chapter 3

- A phase that related with your development process – The development will be based on design
- Design the **actual framework** of your system / product / application / tools / configuration
  - Framework with actual tool/device used
  - Show **the security elements**
- Design the **user interface (UI)** - wireframe
  - Example interface of mobile apps with the details of its function / navigation.

## Sample Interface Design-wireframe

USERNAME

PASSWORD

Forget Password

Username

USERNAME

PASSWORD

Forget Password

Password

Forgot Password ?

- Include the database design (with a real data) – Data dictionary (real table and fields)
  - Entity Relational Diagram (ERD)
  - Relation between table in database – Data Flow Diagram (DFD)
- Security element framework

NO AI GENERATED DIAGRAMS

– How you implement the security element in your application/system

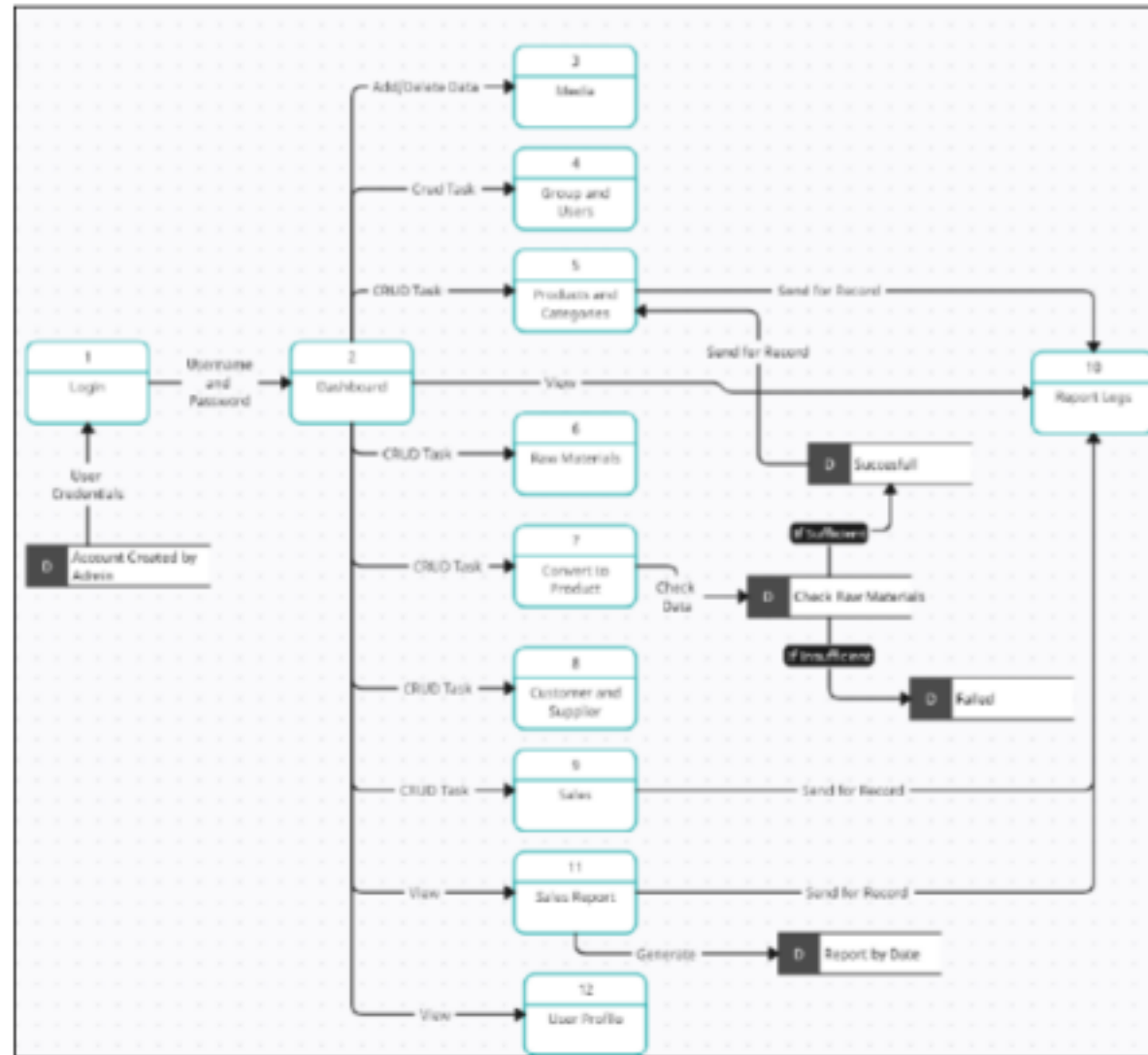
# Example of Data dictionary

# Data Dictionary

Data Dictionary outlining a Online Gaming Service

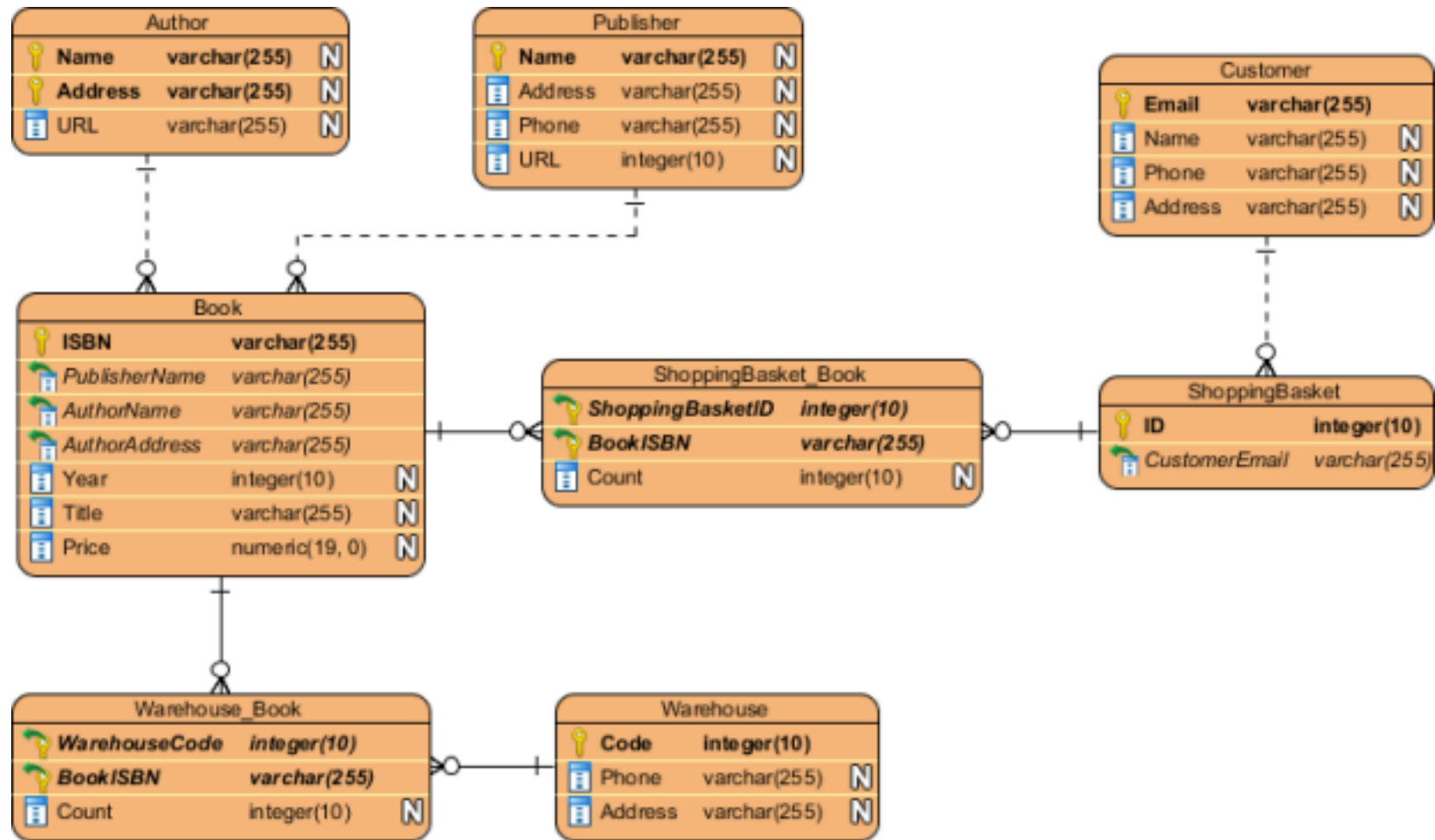
| Data Item            | Data Type                        | Data Format | Number of Bytes for Storage | Size for Display     | Description                  | Example              | Validation                 |
|----------------------|----------------------------------|-------------|-----------------------------|----------------------|------------------------------|----------------------|----------------------------|
| MemberID             | String                           | XNNNNNN     | 7                           | 7                    | Unique Identifier For Member | M123456              |                            |
| First Name           | String                           |             | 25                          | 25                   | First Name of Member         | Scott                |                            |
| Surname              | String                           |             | 25                          | 25                   | Last Name of Member          | Daniels              |                            |
| D.O.B                | Floating Point (Date Format)     | DD/MM/YYYY  | 4                           | 10                   | Birth Date of Member         | 02/04/1990           | Date < Today - 15 years    |
| Platinum Membership? | Boolean                          | X           | 1                           | 1                    | True (T) or False (F)        | T                    |                            |
| Subscription Cost    | Floating Point (Currency Format) | \$NN.NN     | 4                           | 6                    | Cost of Members Subscription | \$27.50              | Cost > 0<br>Cost < \$50.00 |
| Available Game Packs | Array (String)                   |             | 25 * Number of Games        | 25 * Number of Games | Names of Game Packs          | Open World Game Pack |                            |

# DATA FLOW DIAGRAM

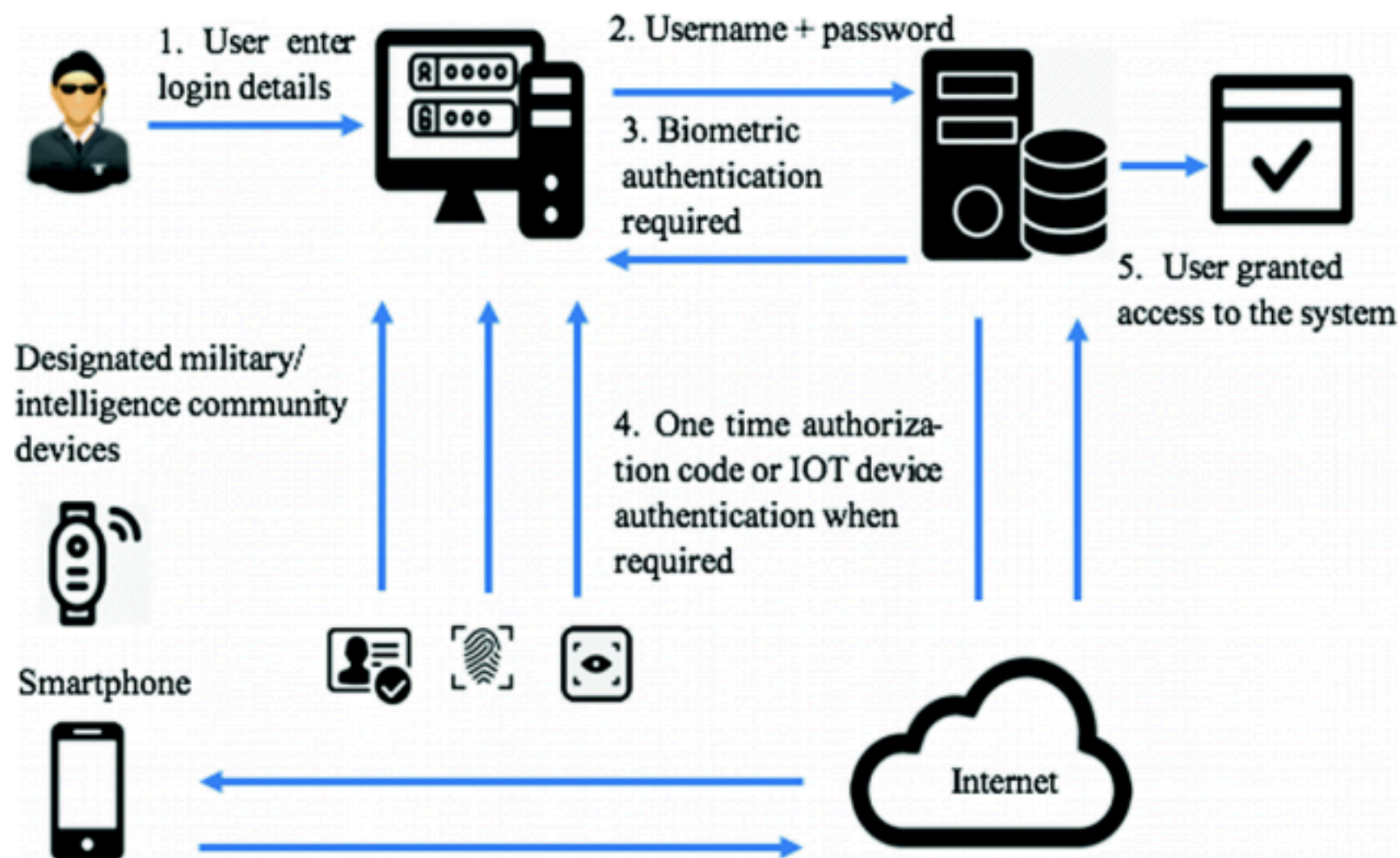


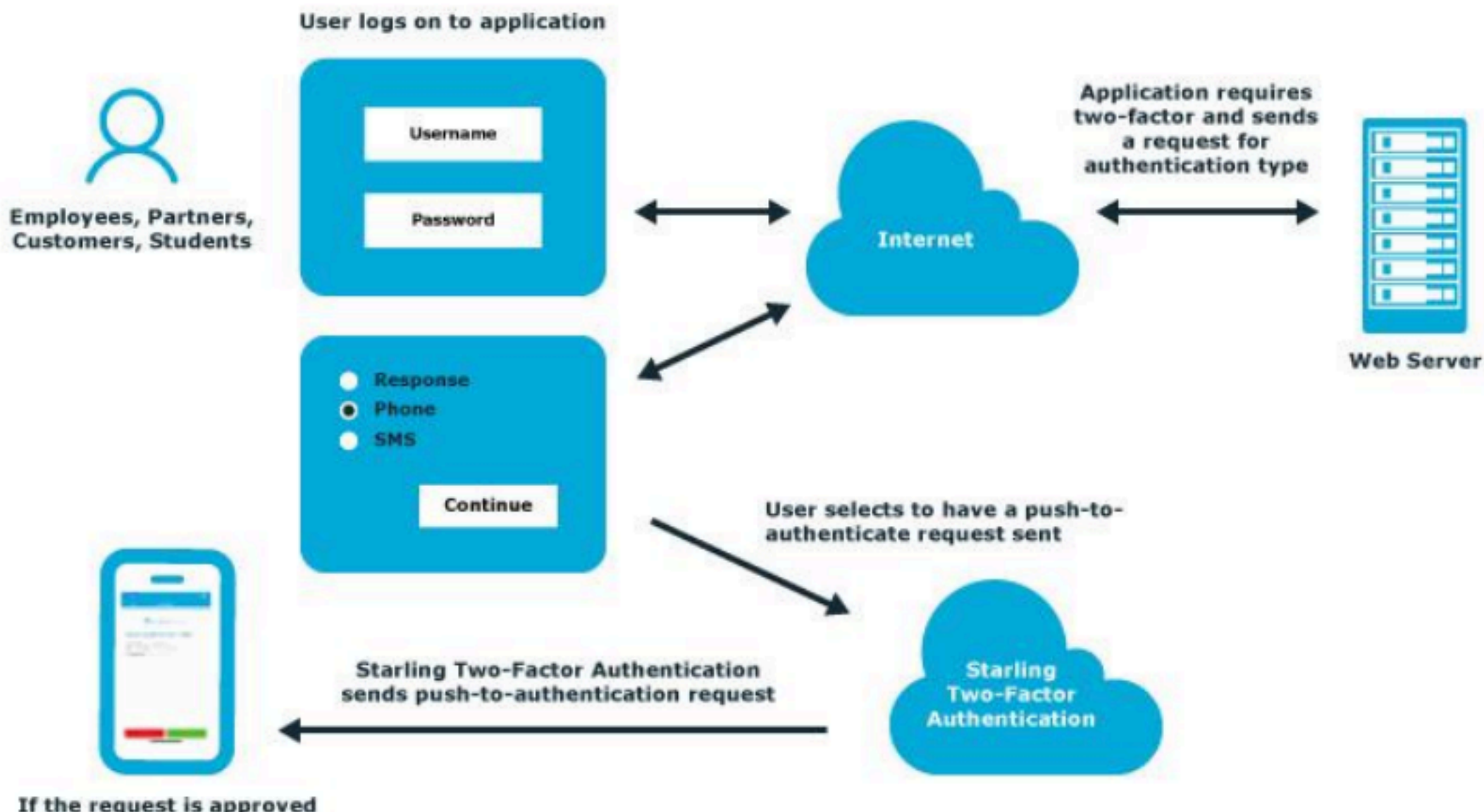


# Example of Entity Relational Diagram (ERD)

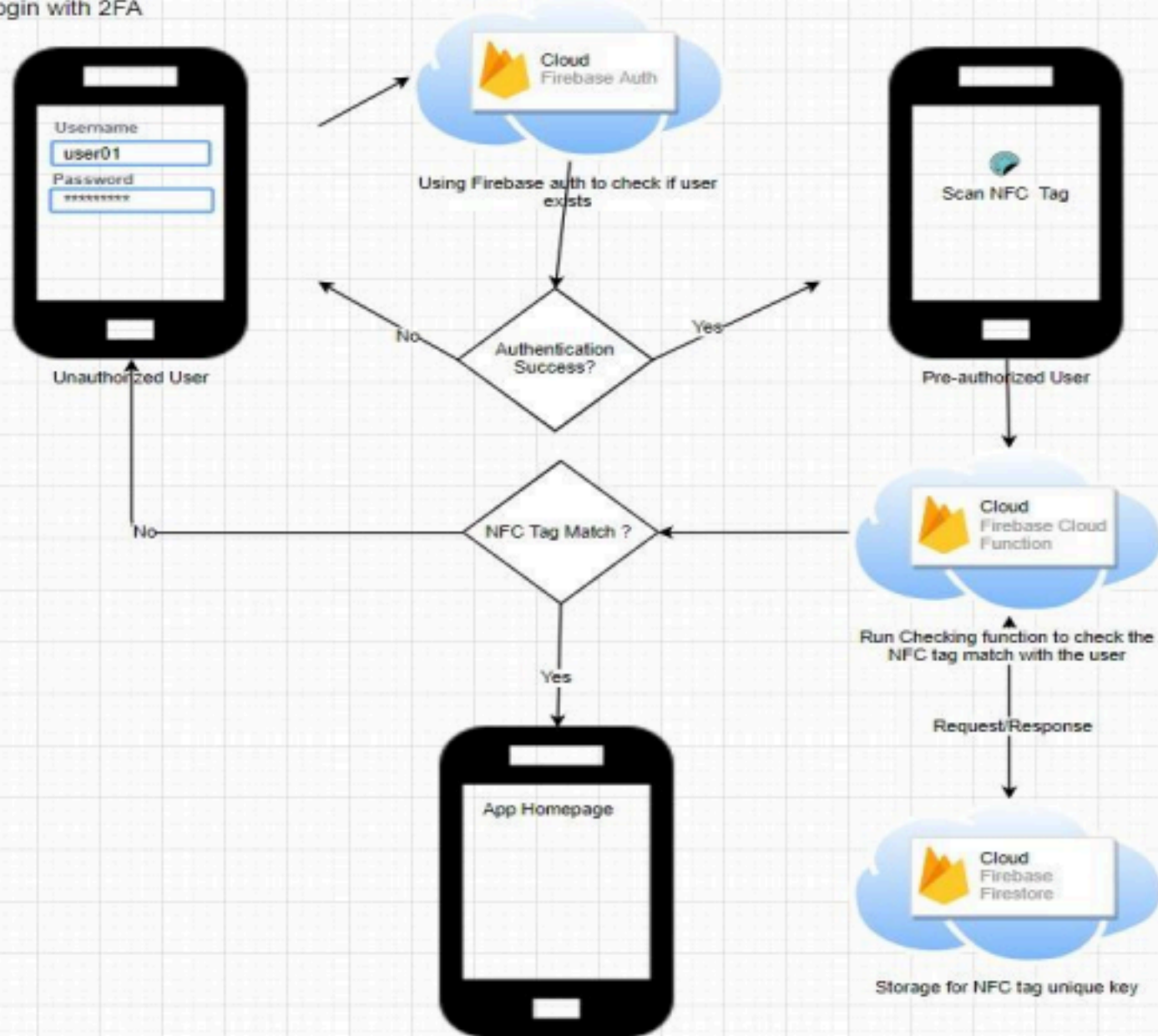


# Example : Security Framework





## Login with 2FA



# Subtopic for chapter 4

## 4 Design

### 4.1 Introduction

### 4.2 Interface design – wireframe

With simple explanation for each figure

### 4.3 Database design

#### 4.3.1 Data dictionary

#### 4.3.2 Data Flow Diagram

#### 4.3.3 Entity Relational Diagram

### 4.4 Security System Framework (FOR CT206 only)

#### 4.4.1 (**Show** the security elements in this framework)

**\*\* This is the recommendation. The subtopic can be change according to your project scope.**

## 4.5 Implementation

- This sub-chapter discuss **the process that you have done** in your project development.
- Explain **how the signification function in the system/application** has been done.
- Explain about the software and hardware (IoT) involve during the implementation process.
- **Show the step** (if tool/network configuration).
- **Show the code snippet** if it is the significant function in your system (System/Application)
- Show the **real user interface / devices (IoT)**

## Example : Algorithm

---

**Algorithm 3:** *ElGamal elliptic curve encryption*

---

**Input:** Parameters field of elliptic curve (  $p$ ,  $E$ ,  $P$ ,  $n$ ), Public key  $Q$ , Plain text  $m$

**Output:** Cipher text  $(C_1, C_2)$

**begin**

1. Represent the message  $m$  as a point  $M$  in  $E(F_p)$
2. Select  $k \in R^{[1, n-1]}$ .
3. Compute  $C_1 = kP$
4. Compute  $C_2 = M + kQ$ .
5. Return  $(C_1, C_2)$

**end**

---

---

**Algorithm 4:** *ElGamal elliptic curve decryption*

---

**Entrada:** Parameters field of elliptic curve (  $p$ ,  $E$ ,  $P$ ,  $n$ ), Private key  $d$ , Cipher text  $(C_1, C_2)$

**Saída:** Plain text  $m$

**início**

1. Compute  $M = C_2 - dC_1$ , and  $m$  from  $M$ .
2. Return ( $m$ ).

**fim**

---

Example : sample coding of security element



#### 4.9.2.2 Password Hashing and Data Protection

User passwords are hashed automatically by using the bcrypt algorithm and then stored in the database. Sensitive data is also hidden from API responses.

```
39     protected $hidden = [  
40         'password',  
41         'remember_token',  
42     ];  
43  
44     /**  
45      * Get the attributes that should be cast.  
46      *  
47      * @return array<string, string>  
48      */  
49     0 references | 0 overrides  
    protected function casts(): array  
50     {  
51         return [  
52             'email_verified_at' => 'datetime',  
53             'password' => 'hashed',  
54             'onboarded_at' => 'datetime',  
55         ];  
56     }
```

Figure 4.84: Code Snippet – Password Hashing and Hidden Fields

Figure 4.84 shows the password hashing and data protection configuration in the CyberSafe

# Subtopic for chapter 4

## 4.5 Implementation

### 4.5.1 Software

#### 4.5.1.2 Programming Language/tool

(Explain in detail **how you use this software/tool in your implementation**) 4.5.2

IoT (**example**)

## 4.6 System Interface

4.6.1 (show **the real interface** of the system for each module)

## 4.7 Significant function (i.e. security element for CT206)

(show the important **code snippet** for the significant function of your system) 4.8

## Conclusion

**\*\* This is the recommendation. The subtopic can be change according to your project scope.**